Cardinal–Ordinal Correlations as Bidirectional Discretization: A RED-Based Operator Account

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Abstract

Cardinals (three) and ordinals (third) are routinely treated as closely related members of the numeral system, yet their semantic contributions diverge: cardinals measure quantity, while ordinals identify positions within an order. This article develops an explicit account of the correlation as a bidirectional discretization architecture in which (i) an ordinalization operator refines a coarse quantity into a structured, addressable progression, and (ii) inverse unification operators collapse refined microstructure back into macro-level ordinalized wholes. The proposal is expressed in a diagnostic methodology (RED: Reverse Entailment Diagnostic) adapted to numerals, testing whether a processual representation can be reified into a discrete countable object (an ordinal) and then reverse-entails back to its cardinal basis under controlled collapse. Formally, ordinalization is modeled as a lexicographic-sum construction R(I,{F_i}) = Σ_{i∈I} F_i, yielding a directed system of refinements whose 'continuity' is processual (deterministic gluing across steps) rather than analytic density. The reverse direction is captured by projection and aggregation maps (collapse c, Fold, group formation G, and GEN) that unify fibers into single indexed objects. Empirically, the framework predicts when (and why) languages allow numeral expressions to shift between quantificational, adjectival, and nominal uses (e.g., the third; came third; the three; three of them), and it derives diagnostics for distinguishing genuine ordinal objecthood from mere contextual ordering.

Keywords

numerals; cardinals; ordinals; ordinalization; type-shifting; nominalization; granularity; RED diagnostic; lexicographic sum; discourse segmentation

1. Introduction

Numeral expressions exhibit a persistent linkage between quantity measurement and ordered position. A cardinal such as three can appear as a quantifier (three books), as a nominal (the three arrived), and as a label in ranking contexts. An ordinal such as third can function as an adjectival modifier (the third book), as a nominal (the third arrived), and as a predicate (he came third). These distributions suggest that cardinal and ordinal meanings are not merely morphologically related; rather, they participate in a systematic semantic architecture supporting controlled shifts between quantity and position.

Existing analyses typically treat cardinals as quantifiers or measure expressions, while ordinals are treated as functions that pick the n-th element of a contextually given ordering. What is less often stated explicitly is a general mechanism that explains why the correlation is productive, stable under syntactic variation, and naturally bidirectional: speakers can both reify progression into ordinal objects and collapse object-like positions back into counting and aggregation.

This paper proposes such a mechanism by aligning a bidirectional diagnostic methodology—RED (Reverse Entailment Diagnostic)—with an order-theoretic operator architecture developed in the accompanying manuscripts. The core claim is that cardinal–ordinal correlation is a specific instance of a broader grammatical capacity: converting a processual representation into discrete addressable units (reification/ordinalization), while retaining a controlled inverse path that collapses or aggregates microstructure back into macro-level objects (unification/collapse).

This paper's novelty is operational: it turns the cardinal–ordinal interface into a bidirectional diagnostic program. By distinguishing modifier, iteration-operator, and reified-NP regimes, and by using RED to test reverse entailment under controlled packaging manipulations, the proposal yields falsifiable predictions about inference strength, coercion costs, and typological variation. The worked case studies (Tables 4–5) illustrate the protocol in practice and provide a template for corpus- and consultant-validated replication.

2. Background: Numerals, ordering, and objecthood

The descriptive distinction between cardinals and ordinals is familiar: cardinals answer how-many questions, while ordinals answer which-in-order questions. Yet the boundary is systematically porous. Ordinals presuppose a domain with at least n elements, yielding a reverse entailment from ordinal statements to cardinal lower bounds. Conversely, cardinals are easily coerced into ordering uses when a context supplies a salient ranking function. Both numeral types can also behave like nominals, suggesting that the grammar provides pathways for numeral expressions to gain (or lose) objecthood.

A second, independently motivated perspective comes from granularity and event individuation. Languages can package processes into countable events or unpack events into subevents. Numeral interconversion is naturally framed as a specialization of this packaging/unpackaging capacity: ordinalization increases internal structure (more granular positioning), while collapse suppresses internal structure (coarser counting and summarization). The manuscripts describe this relation explicitly as bidirectional discretization via refinement and collapse.

2.1 Related work (brief positioning)

The proposal is compatible with standard type-shifting approaches to nominal expressions (e.g., Partee 1987) and with event-based compositionality (e.g., Davidson 1967). What it adds is a specific operator-level bridge between numeral morphology/syntax and granularity control: ordinalization is treated as a constructive refinement operator that builds an ordered object from a coarse index, and the reverse direction is treated as a family of licensed unification operators. This yields diagnostics and predictions that are not stated in purely lexical-derivational treatments of ordinals. For formal semantic analyses of ordinals as rank modifiers that interact with an ordering dimension (including bare ordinals and ordinal–superlative configurations), see Yee (2010) and Alstott (2023). For compositional analyses of cardinals and numeral-containing expressions, see Ionin & Matushansky (2006).

3. RED for numerals: diagnostics for reification and reverse entailment

RED (Reverse Entailment Diagnostic) is a bidirectional methodology that tests whether a representation supports object-like packaging. For numerals, we adapt RED to test whether an ordering process can be reified into a discrete ordinal object, and whether that object can be collapsed back to its cardinal basis under a controlled reverse map.

RED-ORD proceeds in three steps. First, begin with a cardinal counting basis: a domain D and a quantity profile (cardinal information). Second, apply ordinalization, producing an ordered spine and an object-like ordinal term that can be targeted by nominal operators (determiners, case, pronominalization). Third, apply reverse entailment by collapsing the ordinal representation to recover cardinal information (at-least-n presuppositions, frequency distributions over ranks, or stabilized partitions). The diagnostic is successful when both directions are interpretable and compositionally stable.

Crucially, RED-ORD separates ordering-as-pragmatics (contextual ranking without object creation) from ordering-as-grammar (a representation that supports stable reference to positions, ellipsis, and anaphora). Only the latter corresponds to genuine ordinalization in the present framework.

4. Formal architecture: ordinalization, cardinalization, and iteration

4.1 Ordinalization as lexicographic refinement. Let I be a base well-order (an index ordinal) and let $\{F_i : i \in I\}$ be a family of nonempty well-ordered fibers. Ordinalization is defined as the lexicographic sum $R(I,\{F_i\}) = \Sigma_{i\in I} F_i$, where elements of F_i precede those of F_j whenever $i<j$ and each fiber retains its internal order. One application of R replaces each atomic index by an ordered block; iterating R yields a directed system of refinements.

4.2 Coarsening and inverse maps. Refinement is paired with order-respecting collapse maps. A canonical collapse is the projection $c: \Sigma_i F_i \to I$, $c(i,f)=i$, which forgets internal fiber identity. More semantically substantive coarsening is achieved by aggregation/fold operations that merge fiber content into a macro-object (Fold), group formation (G), or generic/habitual unification (GEN), each producing one macro-denotation per index.
4.3 Ordinal assignment and cardinalization as composable maps. In applied settings, it is useful to separate ordinal assignment ord: $S \to O$ (assign items to ordinal labels) from cardinalization card: $O \to N$ (map each ordinal label to the size of its preimage). The composition card∘ ord: $S\to N$ yields per-item counts associated with ordinal bins, and repeated alternation corresponds to functional iteration of these. This perspective makes information loss explicit: cardinalization collapses internal structure, whereas ordinalization introduces it.

4.4 Processual continuity. Although a fixed well-order is discrete, the refinement system induced by iterating R yields an unbroken progression of increasingly specified discrete objects. The manuscripts term this processual continuity and model the corresponding ‘continuous quantity surface’ as a poset of refinements ordered by refinement embeddings, with directed limits capturing asymptotic structure.

Table 1. Operator inventory for cardinal–ordinal correlation

Operation	Formal schema	Linguistic interpretation
Refinement / ordinalization	$R(I,\{F_i\}) = \Sigma_{i\in I} F_i$	Make positions object-like by attaching ordered internal structure.
Collapse / projection	$c: \Sigma_i F_i \to I$; $c(i,f)=i$	Abstract away microstructure; treat many subunits as one indexed

		position.
Fold / aggregation	Fold⊕(F_i)	Package subevents/subparts into a macro-event or macro-individual.
Group formation	G(F_i)	Derive collective individuals; license collective predicates.
Generic / habitual	GEN(F_i,P)	Unify repeated events into a stable generalization anchored at index i.

5. Constructional data: a glossed example set

This section organizes a larger example set by construction type. The intent is diagnostic: each subtype tests either the reification step (ordinal objecthood) or the reverse entailment step (availability of collapse to cardinal information). Examples are illustrative.

5.1 Cardinals in DP (quantificational)

(1) English: three books

three book-PL

‘three books’

(2) English: exactly three books arrived

exactly three book-PL arrive-PST

‘exactly three books arrived’

(3) English: three of the books arrived

three of the book-PL arrive-PST

‘three of the books arrived’

5.2 Cardinals as nominals and labels (reified quantity)

(4) English: the three arrived

the three arrive-PST

'the three arrived'

(5) English: I spoke with the three

I speak-PST with the three

'I spoke with the three (people)'

(6) English: Team Three won

team three win-PST

'Team Three won'

5.3 Ordinals as modifiers (position selection)

(7) English: the third book

the third book

'the third book'

(8) English: a third attempt

a third attempt

'a third attempt'

(9) English: the third time John called

the third time John call-PST

'the third time John called'

5.4 Ordinals as nominals (ordinal objecthood)

(10) English: the third arrived

the third arrive-PST

'the third (person) arrived'

(11) English: I took the third

I take-PST the third

'I took the third (one)'

(12) English: the third of the guests

the third of the guest-PL

‘the third of the guests’

5.5 Ordinals as predicates (rank predication)

(13) English: John came third

John come-PST third

‘John came third’

(14) English: Mary is third

Mary be.PRS third

‘Mary is third (in the ranking)’

5.6 Discourse ordinals (segment indexing)

(15) English: First, John called. Second, he apologized. Third, he left.

first John call-PST second he apologize-PST third he leave-PST

‘First... Second... Third...’

(16) English: Thirdly, the committee rejected the proposal.

thirdly the committee reject-PST the proposal

‘Thirdly, the committee rejected the proposal.’

5.7 Crosslinguistic morphology (illustrative)

(17) Russian: tret′-ij student

third-M.SG student

‘the third student’

(18) Spanish: el tercer libro

the third book

‘the third book’

(19) German: das dritte Buch

the.N third book

‘the third book’

(20) Turkish: üç-üncü öğrenci

three-ORD student

'the third student'

5.8 Diagnostics: objecthood vs pragmatic ordering

Two diagnostics are especially informative. First, nominal behavior: if the ordinal expression supports determiner selection, case marking, pronominal substitution, and ellipsis in the same way as other nominal expressions, the grammar is encoding a stable ordinal object. Second, reverse entailment: if the ordinal entails a cardinal lower bound and supports systematic collapse to counting statements, then the underlying representation must include an ordered spine and an available collapse map. In the operator architecture, nominal behavior corresponds to treating an ordinal as an addressable element or block in a refined order, while reverse entailment corresponds to projection or aggregation that forgets microstructure and recovers cardinal information.

6. From process to ordinal object: the reification path

The forward direction of the proposal is reification: a processual representation is converted into a discrete, countable object with ordinal identity. In the formal model, this is refinement: each coarse index i is supplied with internal structure F_i so that positions become identifiable objects. Because refinement is constructive, it changes the order type unless the target already admits the same consecutive block decomposition. Iterated refinement yields a stable notion of 'continuous specification' in the sense of directed refinement chains. Linguistically, reification is visible whenever an ordinal behaves as an object rather than a mere relation. In the third arrived, the grammar treats third as a nominalized position; it can be modified, referenced anaphorically, and used as the argument of predicates targeting individuals. Within RED-ORD, such behavior motivates the claim that an ordinal denotes (or picks out) a stable element in a refined order created by ordinalization. The manuscripts connect this perspective to event nominalization and segmentation: refinement introduces subevent-level tokens (i,f) that can be quantified over in neo-Davidsonian style. The same formal move explains why ordinal constructions naturally interact with iterative/event-counting environments such as time adverbials (the third time) and discourse enumerations (thirdly).

7. The reverse direction: collapsing discretes into ordinalized wholes

The reverse direction is equally central: languages not only create microstructure but routinely suppress it, collapsing many discretes into a macro-object while preserving ordinal anchoring. This is the coarsening/collapse side of the architecture. The simplest collapse is projection, which maps every element of a fiber to its index. More semantically substantive coarsening is achieved by Fold, group formation, and GEN, which respectively package subparts into macro-events, unify multiple individuals into a collective, and unify repeated events into stable generalizations. This inventory predicts that ordinal identity can remain stable even as microstructure is suppressed. For instance, ranking predicates (came third) are compatible with severe collapse: they preserve the ordinal index while treating within-rank microstructure as irrelevant. Discourse ordinals similarly treat complex segments as single steps, effectively applying unification at the level of discourse fibers. On the diagnostic side, coarsening predicts where reverse entailment should be robust. If a language has access to collapse/unification operations in the numeral domain, ordinals should support systematic recovery of cardinal information (lower-bound entailments, counts per rank, stabilized partitions). If collapse is weak or blocked, ordinal uses may remain tied to pragmatic ordering without stable objecthood.

8. Typological and computational consequences

Typologically, the framework predicts variation in where ordinalization is realized: derivational morphology (dedicated ordinal affixes), functional syntax (ordinal determiners or classifier systems), and discourse-level enumerative operators. Where ordinalization is strongly grammaticalized, ordinal expressions should show robust nominal behavior and stable reverse entailment patterns. Where it is weakly grammaticalized, ordering may be primarily pragmatic and nominal ordinals may require additional support. Computationally, the refinement/collapse architecture supplies an explicit representation for models that must shift granularity on demand. A symbolic implementation represents I and each fiber as arrays and computes R by concatenation with lexicographic keys (i,n), while collapse maps project back to I. A neural implementation can encode coarse indices and fiber sequences with hierarchical attention and an autoencoding objective enforcing order preservation and reconstructability; this aligns directly with the manuscripts' discussion of segmentation and staged processes.

9. Limitations and future work

This paper is deliberately theory-and-diagnostics oriented. It provides an operator account and a structured example set, but does not yet report corpus-based quantitative evaluation. Three empirical risks are particularly salient. First, free adjunction and general nominalization strategies can create apparent ordinal objecthood that is not driven by the numeral system itself; controlled diagnostics are required to separate numeral-specific effects from general packaging mechanisms. Second, ellipsis and anaphora diagnostics can be sensitive to register and genre, potentially inflating or suppressing the availability of nominal ordinals. Third, the proposed refinement/collapse mapping predicts correlations with aspectual packaging and event segmentation, but these correlations require controlled annotation studies and crosslinguistic elicitation.

Future work should therefore (i) build a language-by-language typology that applies RED-ORD to a fixed construction inventory, (ii) run targeted elicitation and corpus studies to test entailment, anaphora, and coercion patterns, and (iii) implement refinement/collapse architectures for numeral-driven segmentation and summarization tasks, comparing them to pragmatic-ordering baselines.

Appendix A: Methods (RED Elicitation and Scoring)

Scope. The RED elicitation protocol is designed to be run with consultant judgments, corpus attestation, or mixed designs. The pilot application reported in this paper is presented in constructed descriptive form to demonstrate the workflow; a submission-ready replication should add consultant metadata and/or corpus counts for each elicitation cell.

Elicitation materials. For each language, elicit constructions C1–C8 using a fixed lexical backbone (e.g., BOOK/LIST/TIME) to reduce discourse confounds. Add minimal variants manipulating definiteness, nominal heads (ONE/PRO), case marking, and agreement where relevant.

Measures. Record (a) acceptability (5-point Likert), (b) preferred packaging regime among {MOD, REIFY, ITER}, and (c) reverse-entailment strength for the C8 prompt on a 3-point scale: 0 = not supported, 1 = context-dependent/weak, 2 = robust/strong.

Ellipsis diagnostics. For C3/C7, elicit an overt-head paraphrase and test whether an overt nominal head is required for argumental distribution, case/agreement, or anaphoric continuation. Classify a language as supporting REIFY if ordinal material can occupy NP slots with stable argument behavior without requiring an overt noun.

Coercion probes. Add forced-coercion contexts that place ITER forms in object-slot environments or force modifier ordinals into argument positions. Substantial acceptability drops (and, where available, processing costs) support a regime-shift coercion cost consistent with the RED architecture.

Reporting standard. For each language, report the grid table plus 3–5 diagnostics motivating regime classification, and provide at least one attestation (corpus or consultant) per construction type when available.

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Crosslinguistic Elicitation Grid

To make the cardinal–ordinal correlation empirically testable and comparable across languages, we provide a fixed elicitation grid. Each language is probed with the same construction types (C1–C8). The grid targets (i) ordinal reification (availability of NP-like ordinals, e.g., ‘the third’), (ii) operator uses (ordinal-as-iteration marker), and (iii) RED reverse entailment from ordinal objecthood back to the cardinal counting base. Glossing conventions (where used): ORD = ordinal marker, CL = classifier, GEN = genitive/possessive linker, DEF = definite, COP = copula.

Table 2. Elicitation grid (C1–C8) for English, Spanish, German, Russian.

Table 3. Elicitation grid (C1–C8) for Turkish, Japanese, Mandarin, Arabic.

Elicitation protocol. For each construction, elicit (i) a neutral declarative, (ii) a contrastive-focus variant, and (iii) a definiteness/nominalization manipulation where available. Record acceptability, preferred reading (modifier vs reified NP vs operator), and inference strength for the RED prompt in C8. These measurements operationalize the discretization and reverse-entailment hypotheses developed in this paper.

Pilot Application: Two Worked Case Studies

To increase publishability, this section applies the elicitation grid as a pilot analysis rather than leaving it as a prospective instrument. We report two worked case studies—English and German—that illustrate (i) how each construction type (C1–C8) maps onto a packaging regime, (ii) how the RED reverse-entailment prompt behaves under modifier vs reified-NP vs iteration-operator regimes, and (iii) which diagnostics most reliably distinguish reified ordinals from ellipsis-only baselines. The pilot uses constructed descriptive examples to make the workflow explicit and replicable; a submission-ready extension can add consultant metadata and/or corpus attestation for each cell.

Case Study 1: English

English distinguishes modifier ordinals (the third book), reified ordinal NPs (the third; the third is missing), and iteration uses (for the third time). The RED view predicts that reverse entailment to minimal cardinality is strongest when the ordinal is packaged as an NP (C3/C7), while iteration uses (C5) decouple ordinals from object cardinality because the counted domain is eventive.

Table 5. Pilot RED grid results for English (C1–C8).

Ellipsis control. A common alternative treats 'the third' as ellipsis for 'the third one'. The RED protocol tests whether ordinal material is licensed in NP positions with stable argument behavior, coordination with other NPs, and anaphoric continuation without requiring an overt nominal head. Where these diagnostics succeed (C3/C7), the protocol classifies the construction as REIFY and predicts stronger reverse entailment than under purely modifier packaging.

Case Study 2: German

German provides an explicit signature of reification because reified ordinals (der/die Dritte) pattern as nominalized adjectives with definite articles and case-marked inflection. Under REIFY (C3/C7) the ordinal behaves like a countable object in the nominal system; under ITER (zum dritten Mal) the counted domain is eventive and the object-cardinality inference appropriately fails. This aligns with corpus-based and structural analyses of substantivized (nominalized) adjectives in German, which motivate a nominal layer over an adjectival base with diagnostics sensitive to degrees of "nouniness" (Fuß 2017).

Table 6. Pilot RED grid results for German (C1–C8).

Cross-case result. The pilot supports a publishable generalization: packaging regime, not ordinal morphology alone, predicts inference behavior. REIFY constructions support the strongest reverse entailment to minimal cardinality; ITER constructions support entailments in an event-count domain; MOD constructions yield domain-relative minimality that is more sensitive to discourse restrictions on the ordering set.

Construction	**Example**	**Packaging regime and diagnostics**	**RED entailment note**
C1	drei Bücher	Cardinal quantifier over NP; baseline counting.	N/A
C2	das dritte Buch	MOD: ordinal adjective with agreement.	Moderate→Strong (domain-relative)
C3	der/die Dritte	REIFY: nominalized adjective functioning as NP; inflects for case/gender.	Strong
C4	A ist der/die Dritte	Copular predication with reified nominal structure available.	Strong (REIFY-supported)
C5	zum dritten Mal	ITER: ordinal counts events/iterations.	None for objects (event-count only)
C6	dritter auf der Liste	MOD/relational: ordinal anchored to explicit list ordering.	Strong (list-relative)
C7	der/die Dritte fehlt	REIFY: ordinal NP in argument position; case alternations available.	Strong
C8	Wenn es einen Dritten gibt, gibt es (mindestens) drei.	RED prompt; robust under REIFY contexts.	Diagnostic item

Construction	**Example**	**Packaging regime and diagnostics**	**RED entailment note**
C1	three books	Cardinal quantifier over NP; baseline counting.	N/A
C2	the third book	MOD: ordinal as rank modifier in an ordering domain.	Moderate→Strong (domain-relative)
C3	the third (one)	REIFY: ordinal packaged as NP (argumental).	Strong
C4	A is third	Predicative ordinal; typically MOD-like but discourse-sensitive.	Moderate (context-dependent)
C5	for the third time	ITER: ordinal counts	None for objects (event-

		events/iterations, not objects.	count only)
C6	third on the list	MOD/relational: ordinal anchored to explicit list ordering.	Strong (list-relative)
C7	the third is missing	REIFY: ordinal NP as subject; argument distribution.	Strong
C8	If there is a third, there are three (minimally).	RED prompt; strongest under REIFY contexts.	Diagnostic item

Distinguishing Predictions

The RED-based operator analysis makes discriminating predictions that differ from a purely lexical-derivational view (ordinals as derived adjectives) and from a purely semantic rank-operator view that treats ordinals as an abstract ordering predicate without grammatical reification. The table below summarizes the highest-yield contrasts and corresponding diagnostic tests.

Table 4. Distinguishing predictions and diagnostic tests (NOT-monograph style).

Related Work and Triangulation

Prior work on numerals spans at least three adjacent traditions: (i) formal semantics of cardinals and their composition, (ii) morphological and lexical-derivational accounts of ordinal formation, and (iii) mathematical treatments of cardinals/ordinals as abstract objects. The RED-based operator view proposed here is compatible with much of this literature but relocates the explanatory center: the cardinal–ordinal interface is not only a matter of suffixation or lexical listing, but a bidirectional packaging relation between continuous ordering procedures and discrete, countable ordinal objects.

On the cardinal side, formal work emphasizes that complex cardinals are compositionally built and interact systematically with NP semantics (e.g., modifier-type analyses of cardinals; Ionin & Matushansky 2006). Standard lexical-derivational accounts treat ordinals as morphologically derived adjectives (English -th, German -te/-ste, Russian adjectival morphology, Turkish -IncI, Japanese dai-, Mandarin dì-, etc.) that contribute a rank/order relation. The RED operator view adds an explicit reification layer: an ordinal exponent can participate in multiple regimes—modifier, reified NP, and iteration operator—with reverse entailment strength predicted to depend on the regime rather than on the exponent alone.

In cognitive neuroscience, both cardinal and ordinal processing implicate parietal systems associated with magnitude and symbolic number, including the intraparietal sulcus. Developmental work suggests that ordinal processing shares substrates with cardinals and may become specialized for symbolic ordinal skills in the left IPS over development (Matejko et al. 2019). More broadly, IPS involvement generalizes to abstract ordinal knowledge beyond strictly numerical sequences (Fias 2007). The present proposal adds a linguistic prediction: morphology and syntactic reification (ordinal-as-NP vs ordinal-as-modifier) should modulate processing costs and neural signatures beyond magnitude alone.

Neuroscience and Computational Applications

Neuroscience. A core claim of the RED approach is that the cardinal–ordinal interface is not only a magnitude computation but a grammar-controlled transition between process-like ordering and object-like reification. This yields neurolinguistic predictions beyond standard number-cognition contrasts. Under canonical modifier uses (C2), ordinal processing should pattern closely with cardinals in parietal magnitude systems. Under reified NP uses (C3/C7), the model predicts additional load from category/packaging operations (nominal licensing, definiteness, NP-slot integration), which should emerge as measurable increases in processing cost and—depending on method—distinct temporal signatures (e.g., increased reanalysis/repair components when a modifier-like ordinal is coerced into an NP regime).

Development and specialization. Evidence that symbolic ordinal processing becomes specialized in the left intraparietal sulcus over development is consistent with a view in which ordinal skills consolidate as a stable cognitive routine (Matejko et al. 2019). On the RED view, that consolidation should be sensitive to the morphosyntax of the learning language: languages that routinely express reified ordinals in NP positions (C3/C7) should encourage earlier stabilization of packaging routines linking ordinal markers to NP-slot behavior, while languages that confine ordinals to modifier-like or predicate-like uses should exhibit different developmental trajectories. This yields testable crosslinguistic predictions for developmental acquisition and for neural specialization profiles.

Computational linguistics. The elicitation grid can be converted into a controlled dataset for evaluating models on (i) ordinal formation, (ii) ordinal reification, and (iii) reverse entailment patterns. For sequence-to-sequence models, a controlled task is bidirectional mapping: cardinal→ordinal and ordinal→cardinal paraphrase/inference (e.g., three→third; dì-sān→sān (minimum inference context)). For language models, surprisal can operationalize RED coercion costs: contexts that force a reified reading (C7) should be more surprising when the model represents ordinals purely as adjectival derivatives than when it represents a distinct reification regime.

Implementation recommendation. A practical formalization introduces two operators: ORD(·) as an ordering/refinement operator and REIFY(·) as a packaging operator that yields a countable ordinal object. The reverse path is captured by a RED-style MINCARD inference rule: REIFY(ORD(n)) licenses the entailment that the underlying cardinality is at least n under the appropriate regime. Evaluation should compare (a) derivational baselines that treat ordinals as adjectival morphology, (b) rank-only baselines that treat ordinals as abstract predicates, and (c) models with explicit packaging and reverse entailment, using the same C1–C8 grid plus targeted minimal-pair contrasts.

Prediction	**RED operator view**	**Lexical-derivational baseline**	**Rank-operator baseline**	**Diagnostic test**
P1 Reified ordinal availability tracks NP diagnostics	Ordinal forms should enter NP slots (C3/C7) when the language provides a general ordinal→NP conversion path (definiteness, nominalizers, classifier support).	NP-like ordinals are treated as idiosyncratic lexical nominalizations, not predicted from general system architecture.	NP uses reduce to implicit ellipsis (‘the third [one]’) without additional grammatical content.	Elicit C3/C7 and test NP-slot behavior (case/determiners/pronominal substitution) crosslinguistically.
P2 Reverse entailment is grammar-sensitive (RED)	RED predicts ordinal→minimal-cardinality inference is strongest under NP-reified readings (C3/C7) and weaker under pure modifier readings (C2).	Entailment is pragmatic; morphology alone does not predict the nominal/modifier asymmetry.	Entailment follows from rank semantics; should not depend on nominal vs adjectival packaging.	Force modifier-only vs NP contexts; measure inference strength (judgments, paraphrase acceptability).
P3 Coercion cost at regime shifts	Switching between operator (iteration) and nominal (reified) regimes imposes processing costs when coerced (e.g., reified use in a language lacking C3).	No special costs expected beyond general lexical selection.	Costs depend on discourse/pragmatics, not on grammatical objecthood.	RT/ERP contrasts for coerced C3/C7 uses vs canonical C2/C5 uses; acceptability gradients.
P4 Morphology–syntax dissociation	The same exponent can realize distinct RED regimes (modifier vs reified NP vs iteration operator), so morphology does not uniquely determine category.	Ordinals are adjectives; category shifts require independent derivation.	Category is semantic; morphology is secondary; predicts weaker syntactic asymmetries.	Test whether ordinal exponents permit determiner/case behavior (C3/C7) independent of adjective agreement (C2).
P5 Typological coupling with	Languages with productive nominalizers/classifiers	No necessary typological coupling	Coupling depends on ordering semantics, not nominalization	Apply C1–C8 grid and correlate C3/C5/C7 availability with independent

nominalization architecture	and event nominalization should expand C3/C5/C7 availability and strengthen RED entailments.	predicted.	resources.	nominalization diagnostics.

Construction	**Turkish**	**Japanese**	**Mandarin**	**Arabic**
C1 Cardinal counting (NP quantification)	üç kitap	san-satsu no hon	sān běn shū	thalāthat-u kutub-in
C2 Ordinal modifier (rank adjective)	üçüncü kitap	dai-san-satsu no hon	dì-sān běn shū	al-kitāb-u ath-thālith-u
C3 Ordinal nominalization (reified ordinal NP)	üçüncü (olan)	dai-san (no mono)	dì-sān (gè)	ath-thālith (minhum)
C4 Copular ordinal predicate (A is third)	A üçüncü-dür	A wa dai-san da	A shì dì-sān	A huwa ath-thālith
C5 Ordinal-as-iteration operator (third time)	üçüncü kez	san-kai-me (de)	dì-sān cì	li-l-marra(t)i ath-thālitha
C6 Ordinal in ordered list (third on the list)	listede üçüncü	risuto de dai-san	zài míngdān shàng dì-sān	ath-thālith fī al-qā'ima
C7 RED reification probe (counting-object compatibility)	üçüncü eksik	dai-san ga nai	dì-sān bù jiàn le	ath-thālith mafqūd
C8 Reverse entailment prompt (de-ordinalization)	Üçüncü varsa, en az üç vardır.	Dai-san ga aru nara, (saitei) san ga aru.	Yǒu dì-sān jiù yǒu (zuìshǎo) sān.	Idhā kāna ath-thālith, fa-hunāka (ʿalā l-aqall) thalātha.

Construction	**English**	**Spanish**	**German**	**Russian**
C1 Cardinal counting (NP quantification)	three books	tres libros	drei Bücher	tri knigi
C2 Ordinal modifier (rank adjective)	the third book	el tercer libro	das dritte Buch	tret'ja kniga
C3 Ordinal nominalization (reified ordinal NP)	the third (one)	el tercero	der/die Dritte	tretij (iz nih)
C4 Copular ordinal predicate (A is third)	A is third	A es tercero	A ist der/die Dritte	A — tretij
C5 Ordinal-as-iteration operator (third time)	for the third time	por tercera vez	zum dritten Mal	v tretij raz

C6 Ordinal in ordered list (third on the list)	third on the list	tercero en la lista	dritter auf der Liste	tretij v spiske
C7 RED reification probe (counting-object compatibility)	the third is missing	falta el tercero	der/die Dritte fehlt	tretij otsutstvuet
C8 Reverse entailment prompt (de-ordinalization)	If there is a third, there are three (minimally).	Si hay un tercero, hay tres (como mínimo).	Wenn es einen Dritten gibt, gibt es (mindestens) drei.	Esli est’ tretij, to est’ (kak minimum) tri.